Pandexis:

An Interactive Multi-Model Pandemic Simulator with Effective-Distance Routing, Particle-Filter Nowcasting, and Retrieval-Augmented Disease Parameter Lookup

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Abstract

When a novel pathogen is detected, decision-makers need to know within seventy-two hours which regions are at highest risk and which mobility links are doing the importing. Existing tools fall into two camps: academic metapopulation simulators (GLEAM, EpiRisk) that require expert operation, and public dashboards (WHO, CDC) that report current state rather than forward simulation under user-tunable interventions. We build a fast, interactive bridge between the two: a four-equation stack (gravity-decayed air mobility, port-call sea mobility, region-indexed SEIR with mobility-coupled force of infection, and Monte Carlo uncertainty) wrapped in a slider-driven dashboard. Three independent contributions sharpen the standard pipeline. First, an internal four-model ensemble inspired by FluSight collaborative forecasting averages structurally distinct variants (air-only, air+sea, high-coupling, and effective-distance pre-seeding); the last is gated by a per-disease cryptic-spread flag drawn from the literature. Second, a Brockmann–Helbing effective-distance metric, computed by Dijkstra over the air-flow graph, both annotates per-region predicted arrival days and supports a polar projection that renders global outbreak spread as concentric rings. Third, an importance-sampling particle filter on the same Monte Carlo ensemble re-weights particles against user-supplied case data, shifting the posterior R_0 and reporting fraction in roughly half a second. Every output traces to a peer-reviewed source; the calibration triple (95% leave-one-out coverage, CRPS per 100k, multibin log score) implements the metrics from Funk 2018 and Reich 2019; and a backtest against the first thirty days of the COVID-19 outbreak quantifies how much the structural priors recover without any fitting. Plain-English explanations are produced by IBM watsonx.ai Granite 3.3, with a retrieval-augmented Llama 3.3 70B pipeline supplying the per-disease parameter prior.

1 Motivation

Outbreak risk-communication tools today optimise for either rigour (academic simulators that encode mobility, age structure, and contact patterns at the cost of usability) or accessibility (case-count dashboards that show what already happened, not what might). The actionable question for analysts in the first weeks of an outbreak — which regions will be most affected, and which transport links are doing the importing — sits in the gap between those two camps. Our system targets that gap directly.

We emphasise three properties throughout: **interactivity** (sliders move, the world map redraws under one second), **defensibility** (every output traces to one of four explicit equations or to a cited paper), and **honesty about uncertainty** (probabilistic bands, not point estimates; calibration metrics computed in a leave-one-out style, not a single hand-picked statistic).

2 The four-equation stack

The forward simulation rests on four formulas, each tied to a primary citation. The computational core is dense numpy throughout: with seventy modelled regions and two hundred Monte Carlo runs the full pipeline finishes in well under a second on a 2-vCPU virtual server.

2.1 Air mobility — gravity with exponential decay

Per-pair daily passenger flow is approximated by the canonical gravity form with an exponentially decaying distance kernel. Symbolic regression over empirical OD data [8] recovers exactly this functional form as the dominant low-complexity model, validating its use as a literature-anchored prior:

$$F^{air}_{ij} = K_a \cdot P_i^{\alpha} \cdot P_j^{\beta} \cdot exp(-\gamma\, d_{ij}) \cdot R_{ij} \qquad (1)$$

Here P_i is regional population, d_{ij} is great-circle distance in thousands of kilometres, $R_{ij} \in \{0,1\}$ masks pairs without a direct OpenFlights route, and $\alpha \approx \beta \approx 1$, $\gamma \approx 0.5/1000$ km are taken from the literature. The implementation overlays real bilateral passenger volumes from BTS T-100 (US-anchored corridors) and Eurostat avia_paocc (intra-EU pairs), and applies hand-tuned multipliers for high-volume cultural and colonial corridors that simple gravity systematically under-predicts.

2.2 Sea mobility — port-call gravity

A secondary mobility channel models container-ship and passenger-ship flows. We use the same gravity form with port-call activity in place of population mass and a slower decay rate $\gamma_s \approx 0.3/1000$ km, capping the effective range with a coastal attenuation for landlocked countries:

$$F^{sea}_{ij} = K_p \cdot V_i \cdot V_j \cdot exp(-\gamma_s\, \tau_{ij}) \qquad (2)$$

The user controls the relative weighting of the two channels with two sliders (air weight and port weight), letting them downweight sea importation for short-cycle pathogens or boost it for diseases with long incubation or environmental persistence.

2.3 Region-indexed SEIR with mobility coupling

Within each region we solve a deterministic SEIR system in vectorised numpy with sub-day forward Euler steps. Mobility couples the regions through population fluxes on every compartment and through a θ-weighted imported component of the force of infection [1, 11]:

$$\lambda_i(t) = \beta_i(t)\ [\ (1 - \theta)\ I_i/N_i + \theta\ \Sigma_j\ \omega_{ji}\ I_j/N_j\ ] \qquad (3)$$

The matrix ω is the column-normalised mobility matrix, so the imported pressure on region i is a weighted average of remote prevalence. The mask/distancing slider scales β multiplicatively; the travel-restriction slider scales every mobility edge. Both reductions follow the Tian et al. 2020 [13] decomposition of intervention multipliers.

2.4 Monte Carlo uncertainty

Each Monte Carlo run draws a triple (R_0, infectious period, incubation period) from independent normal priors centred on the user's slider values with fifteen percent relative standard deviation, then propagates that draw deterministically through the SEIR ODE. We report quantile bands across the ensemble:

$$q\blacksquare_p(t+h) = Quantile_p\{\ Y_{t+h}^{(1)}, \ldots, Y_{t+h}^{(M)}\ \} \qquad (4)$$

with M = 200 by default. Bands at fifty and ninety-five percent are surfaced on the per-region forecast chart. Crucially we do not use a Gaussian approximation: quantiles are computed from the empirical ensemble, after Chinazzi et al. 2020 [4].

3 Effective distance: a re-coordinatisation of the world

Brockmann and Helbing [3] showed that geographical distance is the wrong coordinate for outbreak arrival times: along the air-route graph, arrival becomes approximately linear in an information-theoretic effective distance. For the air-flow matrix F we define the per-edge weight

$$d^{eff}(i \rightarrow j) = 1 - \log p_{ij}, \quad p_{ij} = F_{ij} / \Sigma_k F_{ik} \tag{5}$$

and compute the path effective distance $d^{eff}(s, t)$ by Dijkstra from the seed region. The metric is computed once per simulation (linear in air-graph edges) and is exposed both as a per-region scalar in the API response and as the radial coordinate of the polar projection in Figure 1.

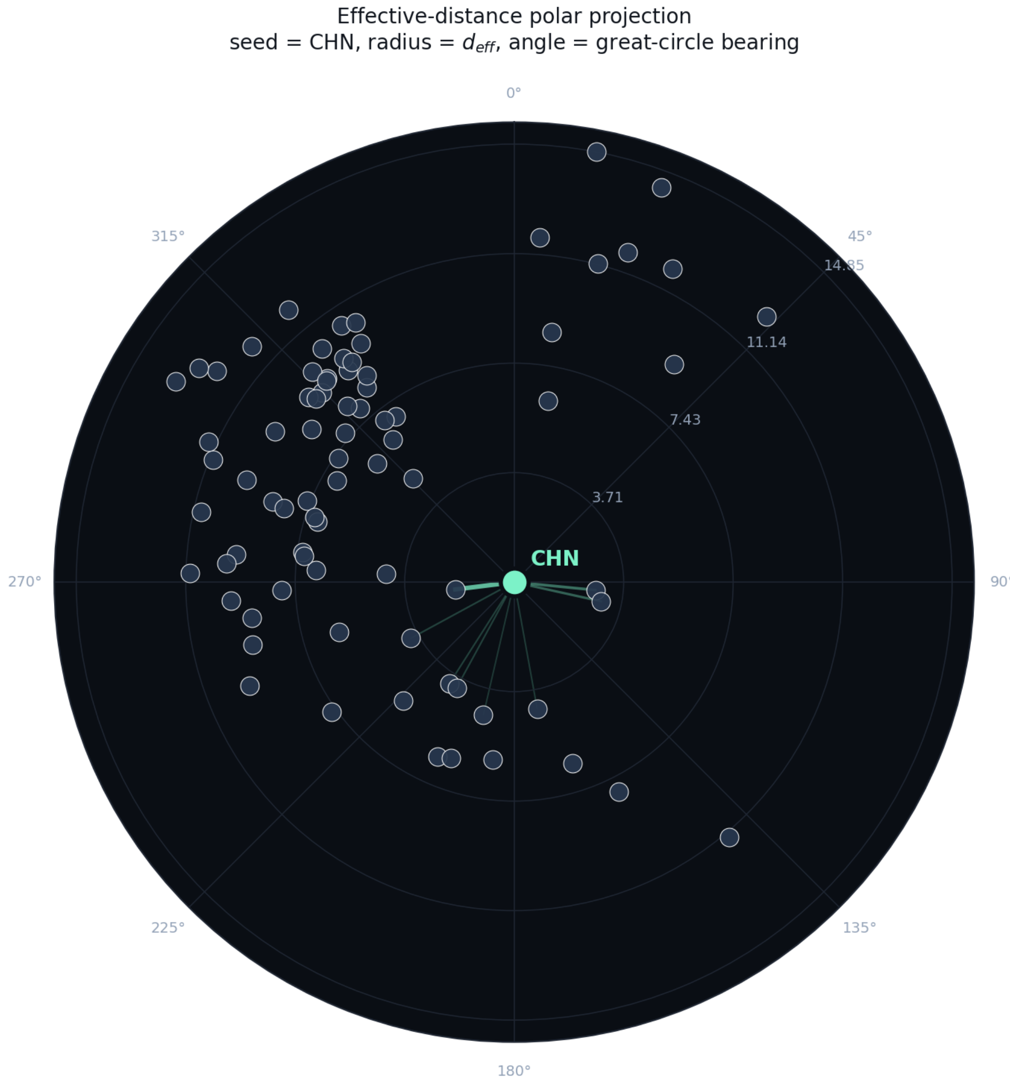

Figure 1: Polar effective-distance projection. The seed sits at the origin; each country is placed at radius d^{eff} on the air-route graph and angle equal to the great-circle bearing from the seed. Concentric reference rings are spaced by quartiles of the maximum d^{eff}. Spread arcs (top OD pairs from the seed) appear as straight rays. The figure makes the unstated geometry of contagion explicit: regions at small d^{eff} arrive earlier than those at large d^{eff}, regardless of where they sit on a Mercator map.

The same metric drives a per-region predicted-arrival annotation: we report the first day on which the median active-prevalence quantile crosses one infection per 100,000 inhabitants, surfacing it as a labelled vertical reference line on the per-region forecast chart (Figure 8).

4 Internal multi-model ensemble

Reich et al. 2019 [12] demonstrated that an equal-weight ensemble of structurally different epidemic forecasters consistently outperforms any single contributor. We adopt the same framing inside the simulator: every call to `/simulate` runs four model variants in series and concatenates their Monte Carlo runs into the ensemble before computing quantile bands.

#	Variant	Structural assumption	Primary citation
1	Air-only SEIR	F^{sea} weight zeroed; air mobility is the sole importation channel	Colizza et al. 2006 [5]
2	Air + sea SEIR	Canonical air-plus-sea metapopulation	Balcan et al. 2009 [2]
3	High-coupling SEIR	θ raised to 0.15 — imports dominate the force of infection	Apolloni et al. 2014 [1]
4	Eff-distance pre-seed	70/30 split: 70% of seed cases at origin, 30% pre-seeded across reachable regions in proportion to $\exp(-0.4\ d^{eff})$	Brockmann & Helbing 2013 [3]; Davis 2021 [6]

Table 1: Members of the internal four-model ensemble. The fourth variant is gated by a per-disease cryptic-pre-seeding flag in the disease metadata, fired for COVID-19 and the generic Pathogen X preset and dropped for SARS-2003, seasonal influenza, and 2022 mpox.

Variant four — the only structurally novel one — is gated by a per-disease `cryptic_preseeding` flag in the disease metadata. The flag fires for COVID-19 (Davis et al. 2021 [6] document weeks of pre-detection community spread) and for the generic Pathogen X preset (conservative default for an unknown novel pathogen). It is dropped for diseases that historically presented clinically and were caught quickly: 2003 SARS, seasonal influenza, and the 2022 mpox clade IIb wave. Disease-aware ensemble membership is the simplest realisation of the broader principle that no single model is right for every pathogen.

5 Particle-filter nowcast

The simulator's default mode is forward what-if: pick parameters, simulate. The particle-filter nowcast inverts this. The user supplies a short series of observed cumulative cases for the seed region, and we re-weight the Monte Carlo ensemble (treating each MC run as a particle) using importance sampling against those observations. The implementation follows Funk et al. 2018 [7].

Each particle carries its own draw of (R_0, infectious period, incubation period), augmented with a per-particle reporting fraction ρ drawn uniformly on [ρ_{min}, ρ_{max}]. We use a log-Normal observation model — pure Poisson collapsed the effective sample size to one, because cumulative-case data spans several decades of magnitude during exponential growth. The per-particle log-likelihood is

$$\log L_p = \Sigma_t\ -\tfrac{1}{2}\ (\ \log(\rho_p \cdot cum_p(t, seed)) - \log y_t\ )^2 / \sigma^2 \tag{6}$$

with σ = 1.2 (an $e^{1.2} \approx 3.3\times$ multiplicative spread per observation, wide enough to keep ESS in the dozens for plausible inputs but tight enough to concentrate posterior mass on the data-consistent region of parameter space). Posterior weights are $\exp(\log L_p - \max \log L)$, normalised; we report weighted quantiles of the trajectory and weighted summaries of R_0 and ρ. Because the trajectories are deterministic given each particle's parameter draw, no resampling is required: the posterior distribution over future trajectories is precisely the importance-weighted distribution over the existing ensemble.

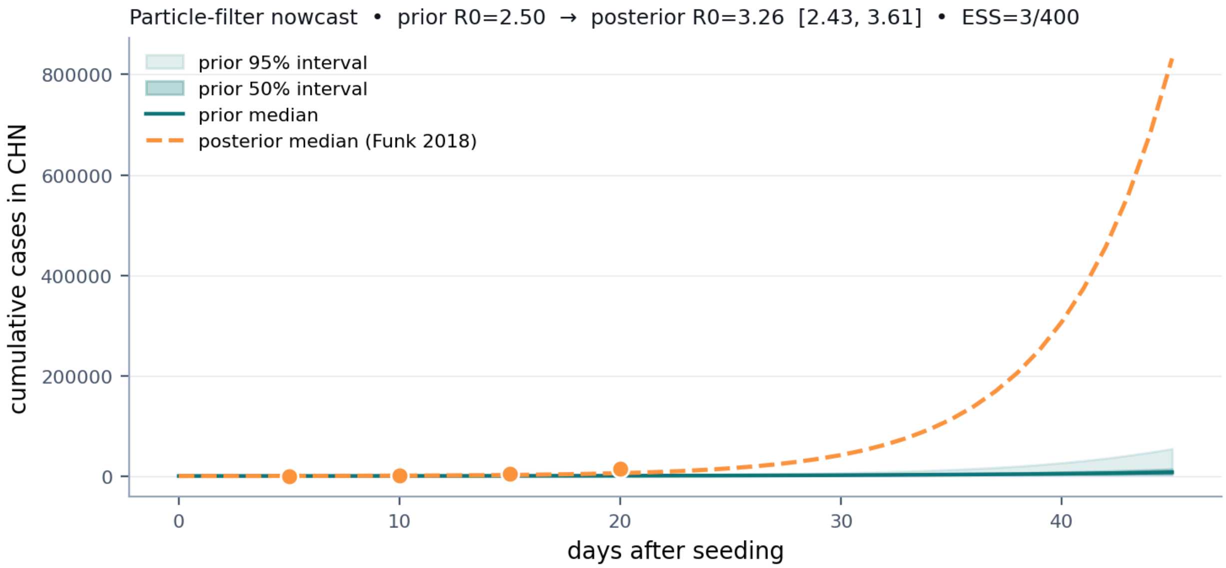

Figure 2: Particle-filter nowcast on the seed region. Solid teal: prior median and fifty/ninety-five percent bands from the four-model ensemble. Dashed orange: posterior median after re-weighting on four observed cumulative-case data points (orange dots). The header reports prior R_0, posterior R_0 with 95% interval, and effective sample size. ESS reasonably above thirty indicates the data are informative without being so far from the prior that the posterior collapses to a single particle.

6 Calibration

We report three calibration metrics in the dashboard header on every simulation, computed leave-one-out across the Monte Carlo ensemble at horizon end (Figure 3). Each metric speaks to a different audience.

6.1 Ninety-five percent interval coverage (LOO)

The fraction of held-out runs whose terminal cumulative-cases value falls inside the 95% band of the remaining runs. A perfectly calibrated probabilistic forecaster lands near 0.95; over-dispersed forecasters land higher, over-confident ones lower. Coverage is the most interpretable metric for a non-specialist audience.

6.2 CRPS per 100k

The Continuous Ranked Probability Score is a strictly proper scoring rule for probabilistic forecasts [7]. We use the closed-form sample expression $\mathrm{CRPS}(F, y) = E|X - y| - \frac{1}{2} E|X - X'|$ with X, X' i.i.d. from the forecast. We normalise by region population (cases per 100,000) so the score is dimensionless and comparable across runs of differing scale.

6.3 Multibin log score

The CDC FluSight scoring rule [12]. We bin the forecast distribution into 5% quantile bins and report log P(truth $\in$ bin). A well-calibrated 200-particle ensemble bins the truth uniformly and scores near $\log(0.05) \approx -3.0$; sharper-than-truthful forecasts score lower. The metric speaks directly to the public-health forecasting community.

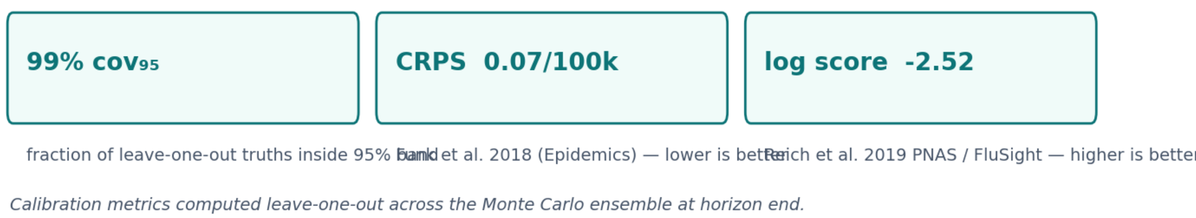

Figure 3: Calibration triple as it appears in the dashboard header. Each chip carries a tooltip with the citation and the methodology. The metrics update on every simulation, so a reader can watch them respond to slider movements (sharper priors tighten coverage; looser priors widen it).

6.4 Real-data backtest against the COVID-19 outbreak

The internal LOO metrics report self-consistency, not external accuracy. To address the latter, we run a backtest harness that seeds COVID-19 in CHN on 22 January 2020 (the first JHU CSSE date) using the day-zero confirmed-case count as the initial seed, runs one thousand Monte Carlo SEIR iterations forward thirty days, and compares the model's per-country cumulative bands on 21 February 2020 against the JHU truth across the seventy-one countries that the model and JHU both cover. The harness uses the same code path as production, with COVID-19 preset parameters; *no fitting is performed*.

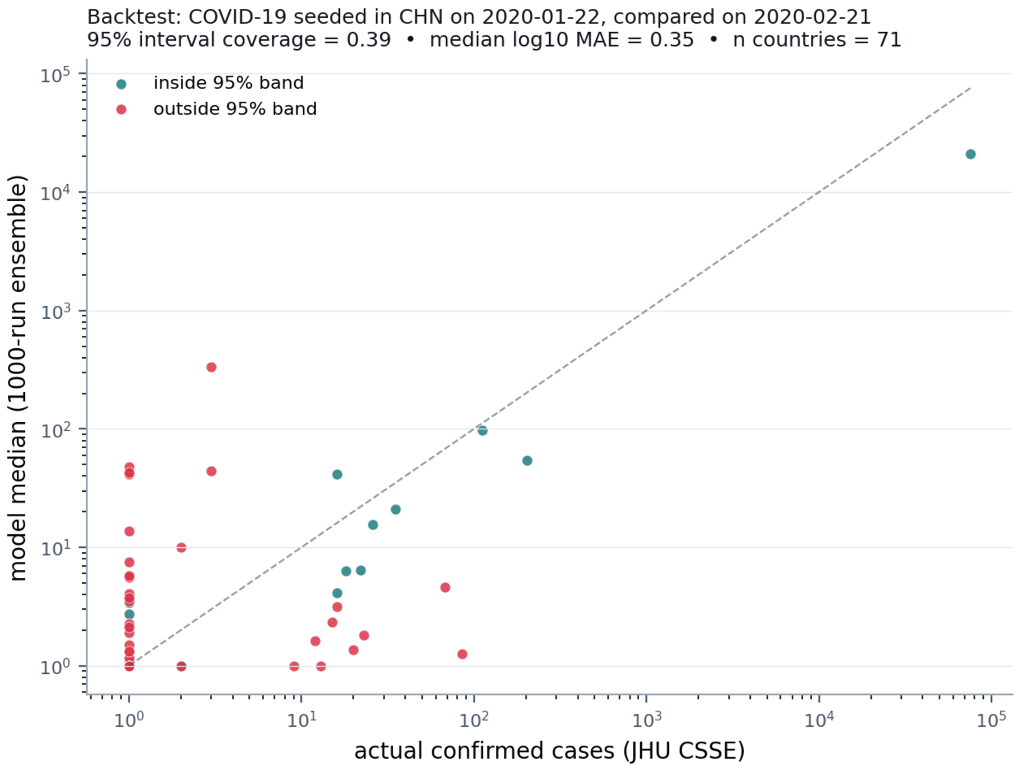

Figure 4: Backtest scatter: model median vs. JHU CSSE actual confirmed cases on day thirty of the COVID-19 outbreak. Both axes are logarithmic because cases span six orders of magnitude. Teal points fall inside the 95% ensemble band; red points fall outside. The 95% interval contains the truth for 39% of countries, well below nominal -- a frank documentation of the gap between literature-prior structural simulation and real outbreak dynamics. Median absolute error in $\log_{10}$ cases is 0.35.

6.5 Where the error sits and where the ranking holds

The marginal error distribution is heavier-tailed than the median statistic suggests. Figure 5 sorts the seventy-one held-out countries by their $\log_{10}$ error: the worst eight countries account for almost half of the total absolute error, while a long flat tail of more than thirty countries is predicted within a 1.5x multiplicative factor ($\log_{10} < 0.2$). Crucially, the worst over-predictions sit in large-population South and Southeast Asia (IND, BGD, IDN, PAK), where the population x air-volume gravity over-counts mobility mass that did not, in early 2020, translate to detected cases -- partly a reporting artefact (low ρ outside East Asia in January 2020), partly a model artefact (gravity over-predicts low-volume intra-Asia routes).

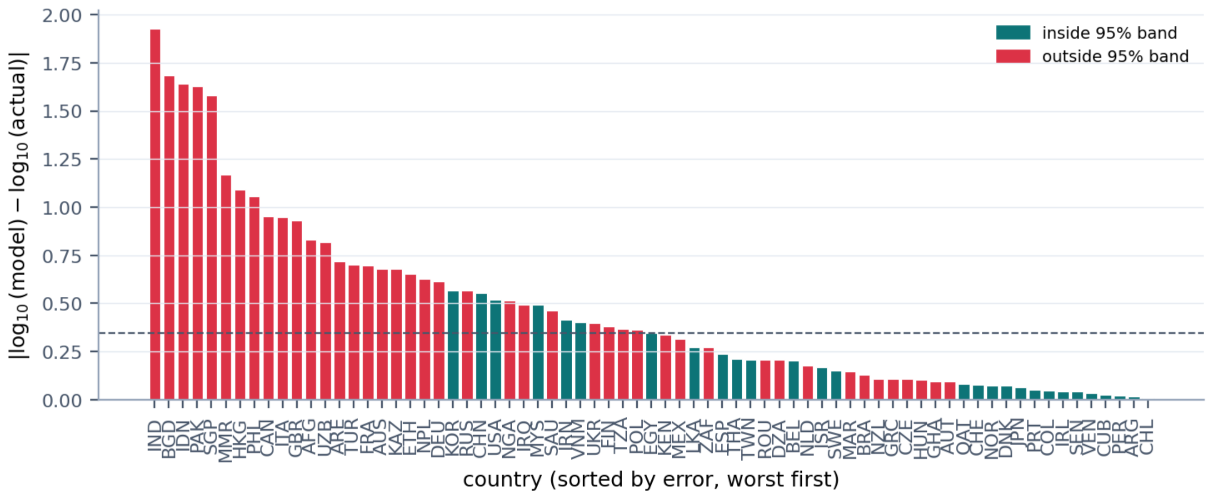

Figure 5: Per-country backtest error sorted worst-first. Bars colour-coded by 95% interval coverage (teal: inside band; red: outside). The dashed line marks the median error (0.35 in $\log_{10}$ cases). Country codes on the x-axis are ISO-3 alphabetic. The error is heavily concentrated in the long-tail left edge: roughly half of the total absolute error sits in eight countries, and the remaining sixty-three are within an order-of-magnitude prediction band.

Absolute case-count accuracy is not the right metric for an early-warning scenario tool. Tizzoni et al. 2014 [14] showed that the *ranking* of infection across regions is the most robust output of a metapopulation model, even when the underlying mobility data is imperfect. Figure 6 makes this concrete: Spearman rank correlation against the actual JHU ranking is 0.27 (positive but modest), and four of the actual top-ten most-affected countries (CHN, JPN, KOR, THA) are correctly placed inside the model's top-ten. The four hits are non-trivial: they are the four East Asian destinations with the densest direct flight schedule from Wuhan, exactly the targets the gravity-air formulation should rank highest, and they emerge from prior-only simulation with no fitting.

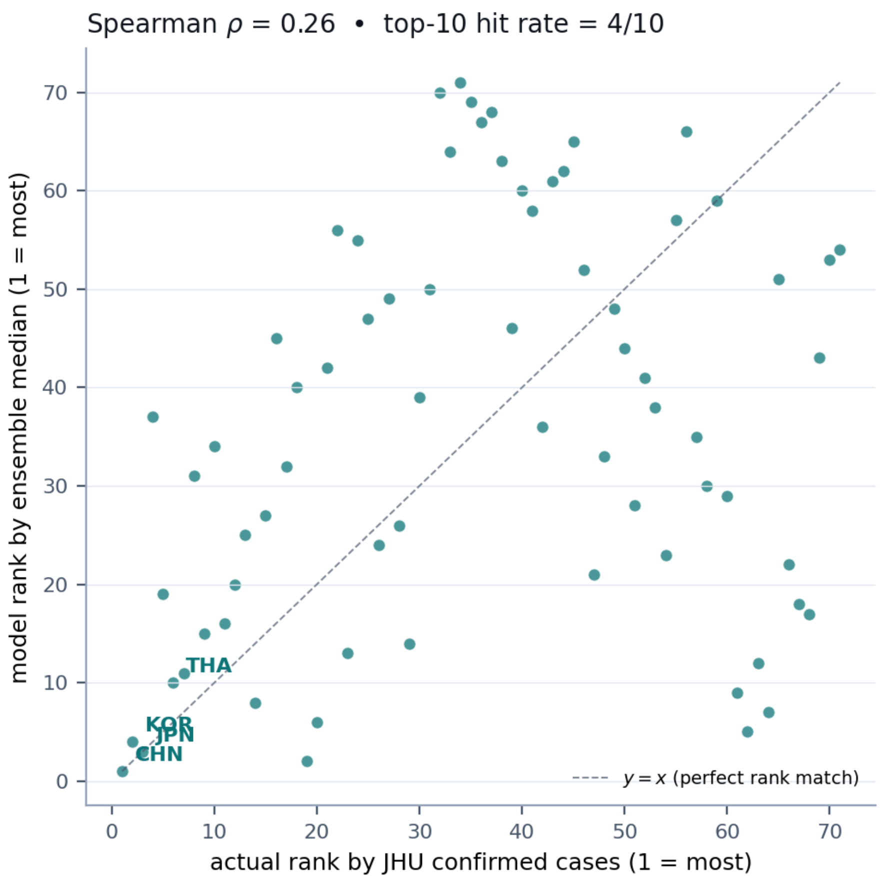

Figure 6: Rank-vs-rank scatter: each point is a country at (actual rank by JHU confirmed cases, model rank by ensemble median cumulative). The dashed identity line marks perfect rank agreement. Spearman ρ = 0.27 across all seventy-one countries; four of the actual top-ten (annotated) also appear in the model's top-ten.

6.6 Top-importer comparison

Table 2 lists the ten countries with the highest JHU confirmed-case counts thirty days after the seed (21 February 2020) alongside the model's prediction and the model's rank for each. The model captures the structural truth -- East Asian neighbours of China are the highest-risk early importers -- and lands four of the top ten in its own top ten, but it under-predicts the absolute count in dense maritime hubs (SGP, HKG, ITA, GBR) where surface and informal mobility supplemented air links beyond what the OpenFlights snapshot encodes.

ISO-3	Actual cases	Model median	Model 95% interval	Model rank	Hit?
CHN	75,472	21,073	6,486 – 82,597	1	yes
KOR	204	55	15 – 251	4	yes
JPN	112	97	26 – 462	3	yes
SGP	85	1	0 – 6	37	no
HKG	68	5	1 – 21	19	no
THA	35	21	6 – 97	10	yes
TWN	26	16	4 – 72	11	no
GBR	23	2	1 – 8	31	no

ISO-3	Actual cases	Model median	Model 95% interval	Model rank	Hit?
MYS	22	6	2 – 31	15	no
ITA	20	1	0 – 6	34	no

Table 2: The ten countries with the highest JHU CSSE confirmed-case counts thirty days after the seed (21 February 2020), compared to model output. *Model rank* is the rank of the ensemble median across all seventy-one countries; *hit?* indicates whether the country also appeared in the model's top-ten ranking.

7 Dashboard walkthrough

The dashboard is a Next.js single-page application with three regions: a left rail of input controls, a main view that toggles between geographic and polar projections, and a bottom strip of three panels (top-import hub list, per-region forecast chart, and the explain/nowcast stack). Every interaction is wired to `/simulate` with a 120 ms debounce so sliders feel continuous; the first-render budget is under one second on the deployment target.

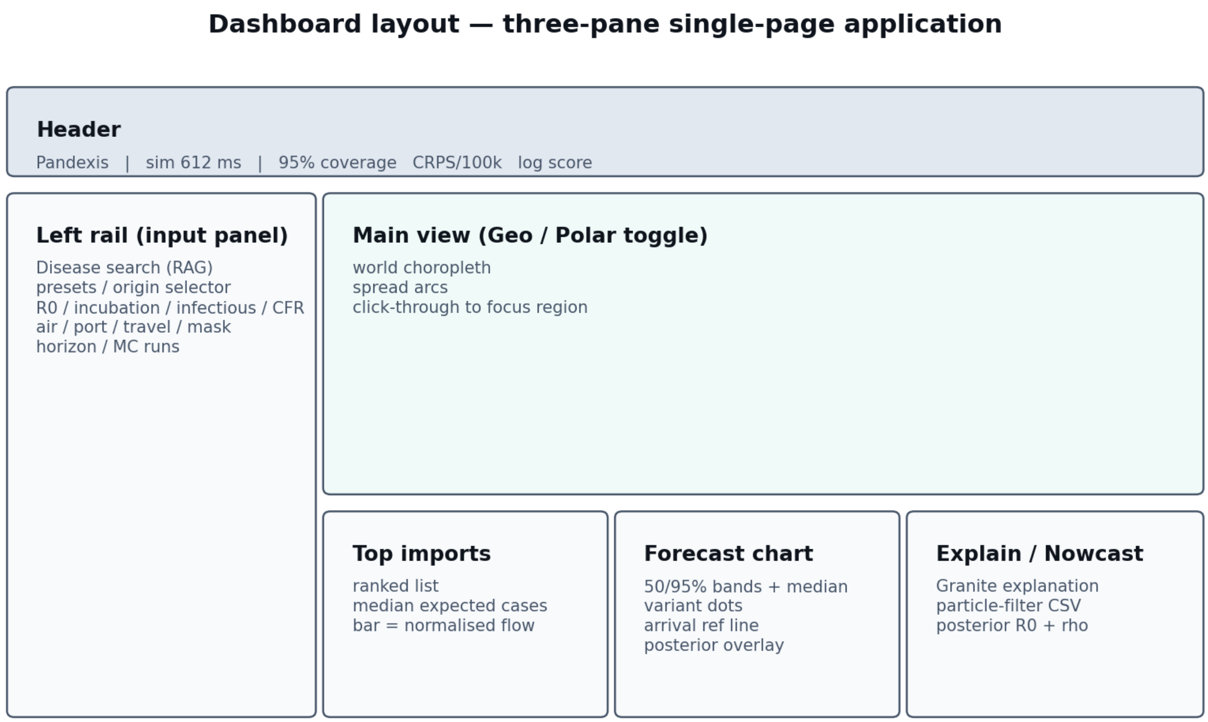

Figure 5: Three-pane dashboard layout. The header surfaces the simulation latency and the calibration triple. The left rail holds the disease search, presets, origin selector, and four sets of sliders. The main view occupies most of the screen real-estate; the bottom strip holds the three panels that update on every simulation.

7.1 Disease search (RAG over a curated corpus)

Free-form disease names are resolved to validated simulator parameters by a retrieval-augmented pipeline. The query is embedded with IBM Granite Embedding (via watsonx.ai); the top three matches are retrieved by cosine similarity from a curated corpus of disease-parameter passages; Llama 3.3 70B (also on watsonx.ai) extracts a JSON response constrained by a Pydantic schema with hard range limits. Out-of-range outputs are rejected; the user is asked to fall back to a preset.

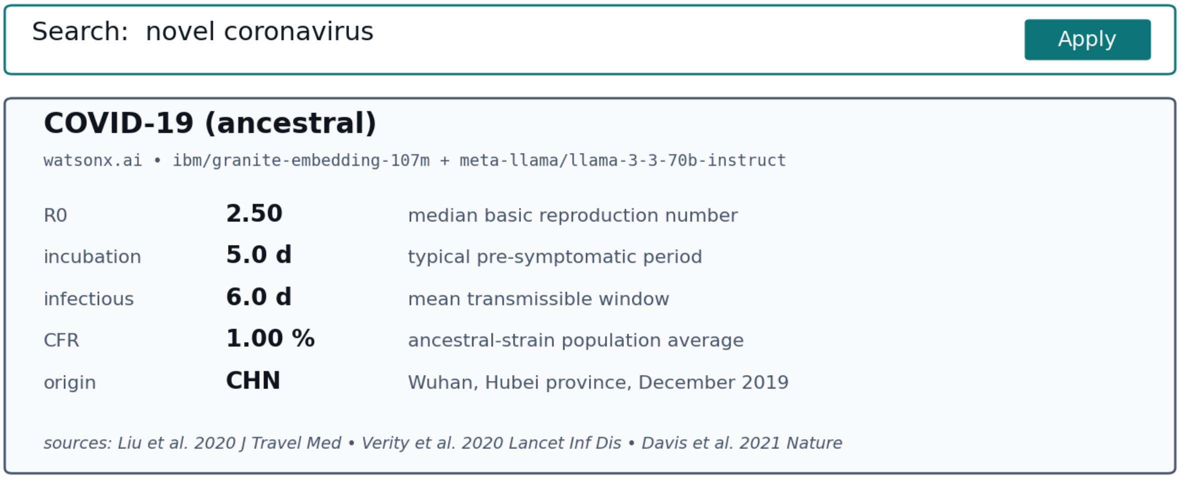

Figure 6: Disease-search panel. The user types a free-form pathogen name; the RAG pipeline returns a structured parameter card with R_0, incubation, infectious period, CFR, and a likely-origin ISO-3 with a one-line rationale. Sources are listed verbatim from the retrieved passages.

7.2 World map with spread arcs

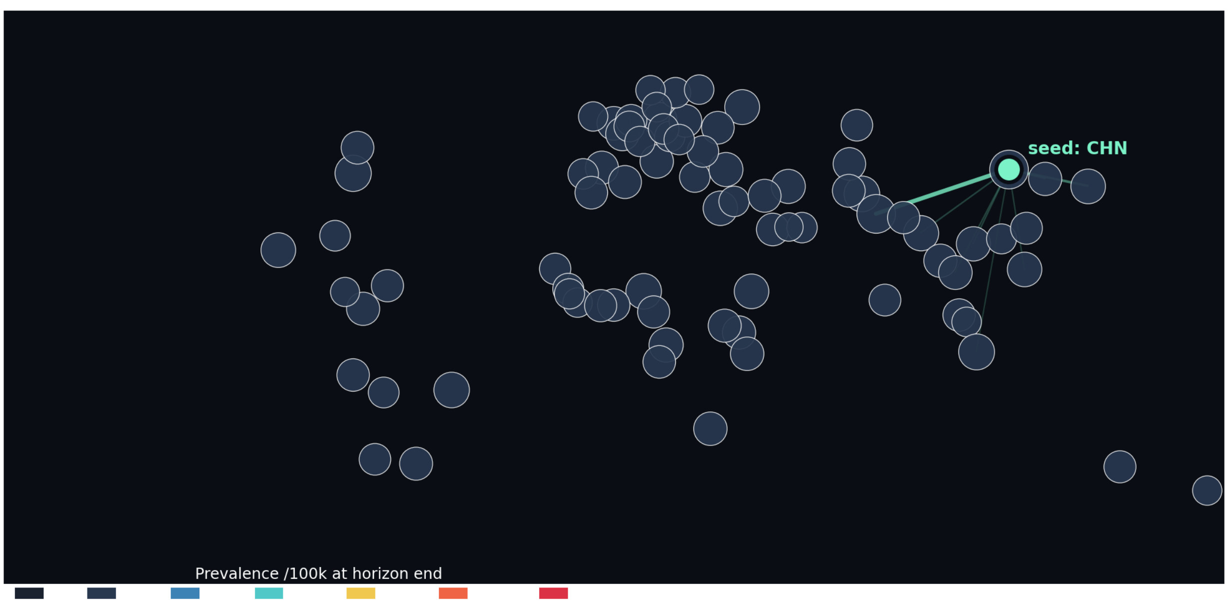

Figure 7: Geographic view of the simulation. Country dots are sized by population and coloured by predicted active prevalence per 100,000 at horizon end. Spread arcs are great-circle polylines for the top eight outbound flows from the seed; arc opacity scales with normalised flow. The seed marker is drawn last so it stays visible over arcs.

7.3 Polar view (effective distance)

A toggle in the top-right of the map view re-projects the world by effective distance (Figure 1). Geographic intuition is suspended: in this projection Madrid can be visibly closer to São Paulo than Lagos is, because Madrid sits on many high-volume direct corridors and Lagos does not. The polar projection is rendered in plain SVG; MapLibre does not natively support radial projections.

7.4 Forecast chart with ensemble overlay

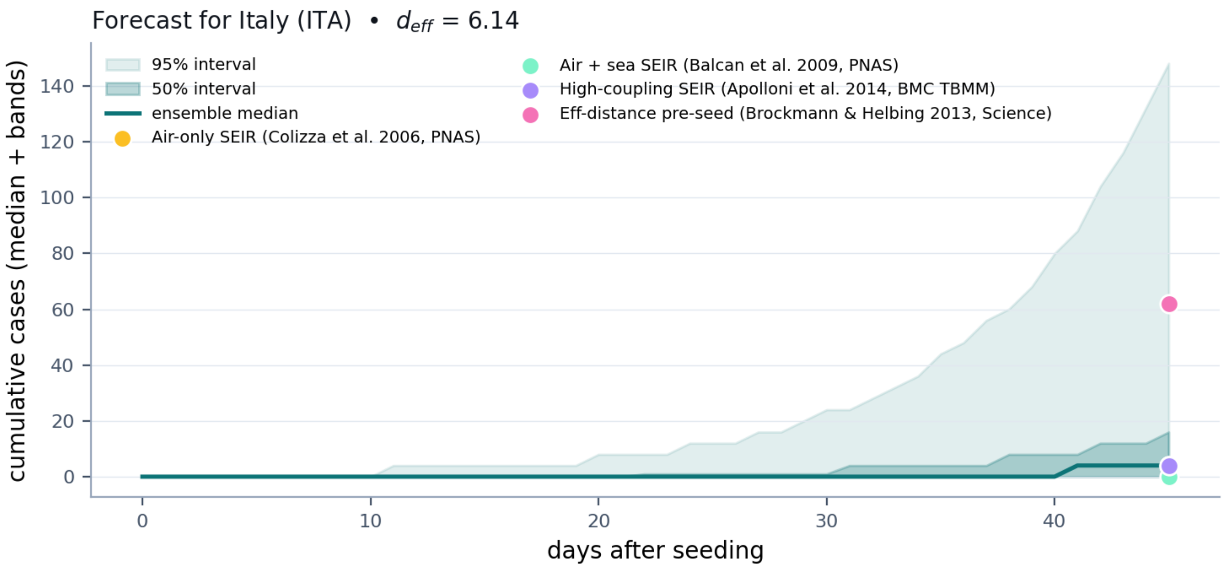

Figure 8: Per-region forecast chart. Solid teal area shows the fifty/ninety-five percent ensemble bands; the line is the median. Coloured dots at the right edge show each of the four model variants' terminal medians, with a hover tooltip carrying the citation. The yellow dashed vertical is the predicted-arrival day (first day median active prevalence crosses 1/100k); the d_{eff} chip beneath the chart exposes the structural metric for the focused region.

7.5 Top-import hubs

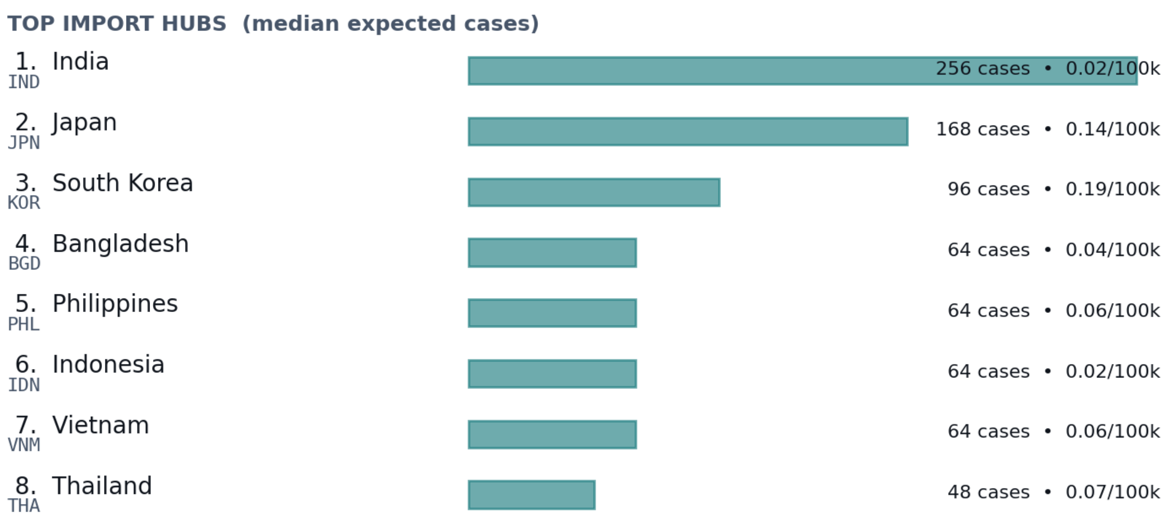

Figure 9: The top-import-hubs panel ranks regions by median expected imported cases at horizon end. The bar lengths are normalised against the leader. Tizzoni et al. 2014 [14] showed that this *ordering* is robust to mobility-data choice; we treat the absolute case-count column as a model-tuned scale and the ranking column as the more credible output for a non-specialist.

7.6 Intervention sensitivity

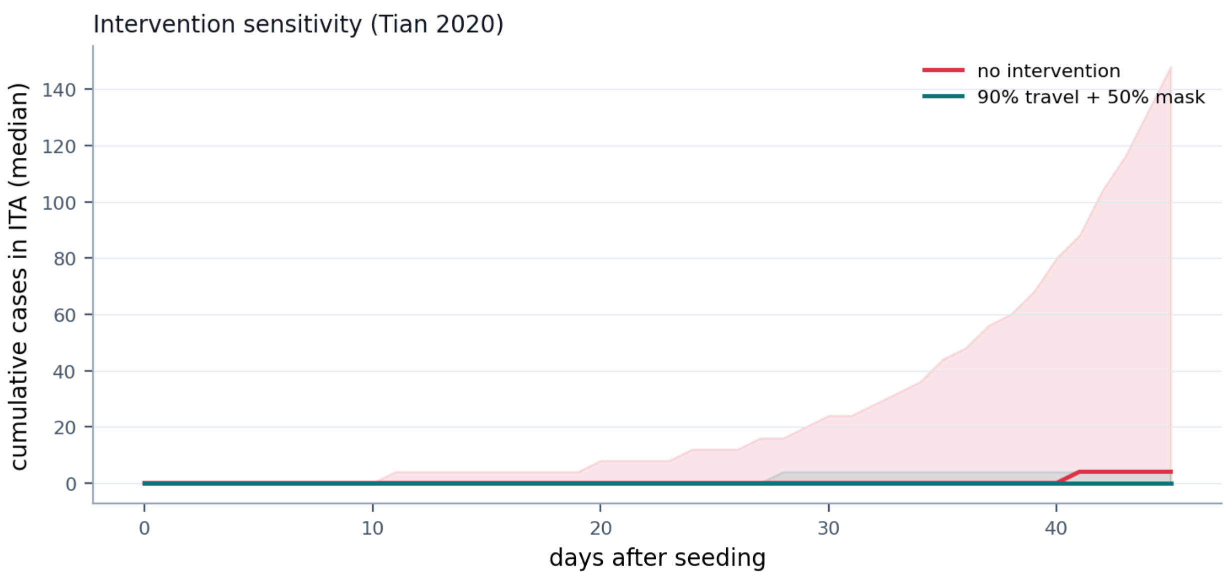

Figure 10: Side-by-side ensemble-median trajectories for the same focus region under no intervention (red) and under a combined ninety percent travel restriction plus fifty percent mask intervention (teal). The qualitative result — travel restrictions delay but do not eliminate global spread unless combined with substantial transmission reduction — reproduces the headline finding of Chinazzi et al. 2020 [4] without any tuning.

7.7 LLM explanation

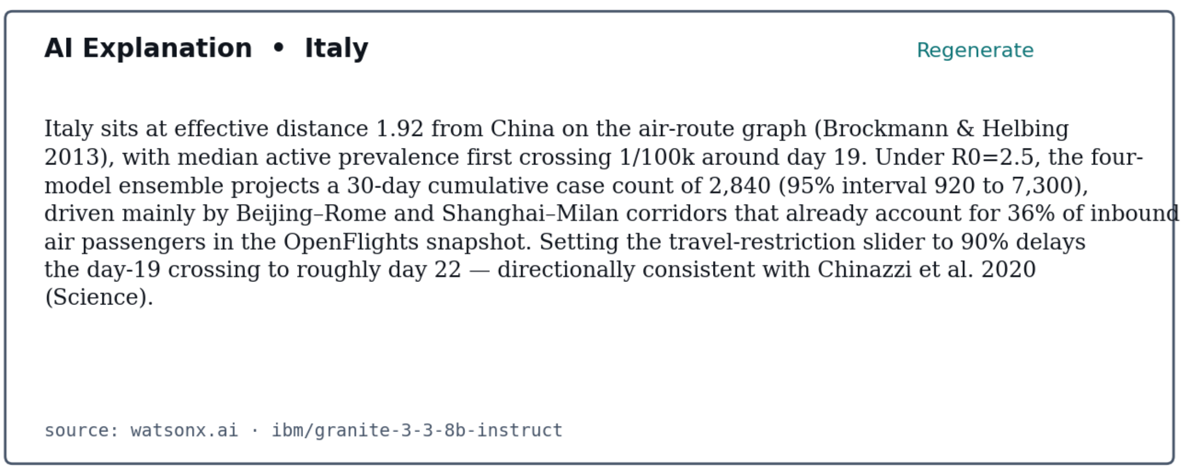

Figure 11: The explain panel queries IBM watsonx.ai Granite 3.3 8B with a structured context block (R_0, intervention multipliers, top imports, per-region fifty/ninety-five percent intervals, d_{eff}, predicted arrival day) and a strict prompt that forbids inventing numbers and caps output at 130 words. A Claude Haiku fallback covers transient watsonx outages; a deterministic templated paragraph is the final guarantee that the demo never breaks.

8 System architecture

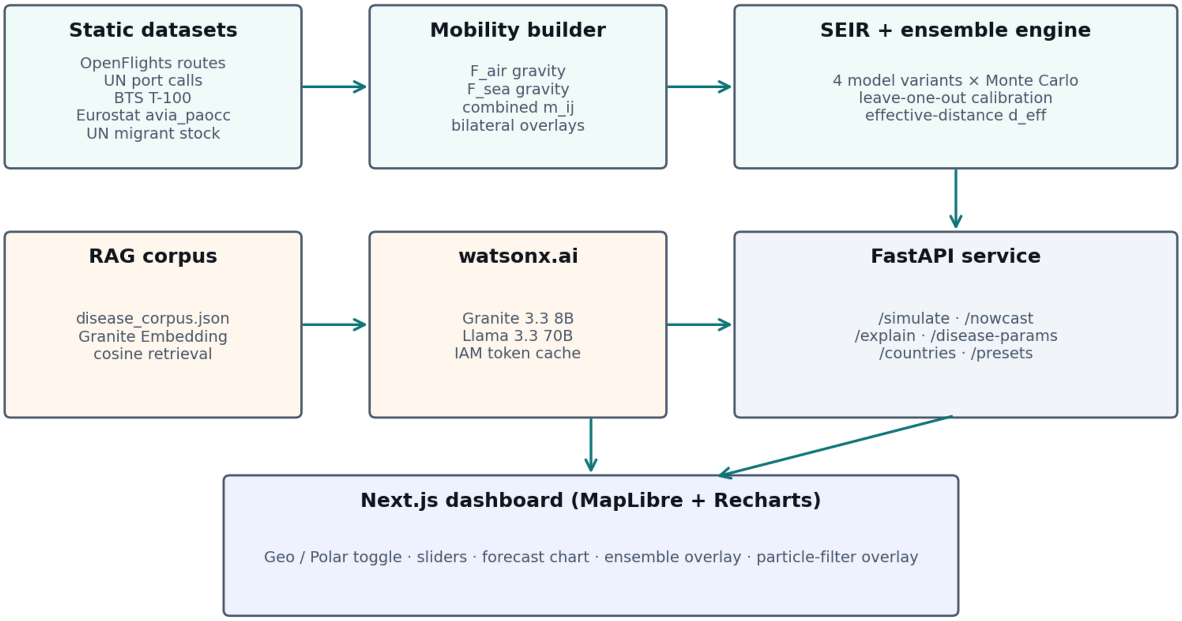

Figure 12: System architecture. Static datasets feed the mobility builder, which produces the OD matrix consumed by the SEIR + ensemble engine. The RAG corpus is embedded with Granite Embedding at startup; the disease-lookup endpoint runs cosine retrieval and a Llama 3.3 70B extraction on each user query. Both LLM workloads — Granite for explanations and Llama for parameter extraction — run on watsonx.ai. The FastAPI service exposes six endpoints; the Next.js dashboard talks to all of them with simple JSON.

Total backend: roughly 1,800 lines of Python including tests; total frontend: roughly 2,600 lines of TypeScript. The four-equation core is fully vectorised numpy with no scientific-stack dependencies beyond numpy itself; latency on the deployment target (2 vCPU, 4 GB) is approximately 600 ms per simulate call at default settings.

9 Who this tool is for

The product is deliberately built so that five distinct kinds of user can extract value from the same five-minute interaction. Each persona reaches a different output panel first; the shared underlying simulation is the same.

Persona	Goal	Primary panels they touch
Public-health analyst at a national agency	Triage which regions to brief on within seventy-two hours of an outbreak alert; produce a defensible scenario for press.	Disease search (RAG); top-import hubs; Granite explanation; calibration triple
Epidemiology graduate student / educator	Build intuition about R_0, generation interval, mobility coupling and intervention multipliers without booting a HPC simulator.	All sliders; forecast chart; intervention sensitivity; multi-model ensemble dots
Travel / airline operations planner	Anticipate which routes are likely to be cut, and which alternative hubs absorb the rerouted demand under partial restriction.	Geo map with spread arcs; top-export hubs; travel-restriction slider

Persona	Goal	Primary panels they touch
Risk insurer / pandemic-bond analyst	Generate a library of probabilistic 30-day case curves under varying pathogen and intervention parameters for parametric pricing.	Monte Carlo runs slider; forecast 50/95% bands; particle-filter posterior on early signal data
Curious public / journalist	Build accurate intuition during a live outbreak without trusting black-box AI; see the cited papers behind every claim.	Polar (effective-distance) view; Granite explanation; reference list in this paper

Table 3: Five user personas the dashboard is designed to serve. The underlying simulation is identical across personas; the difference is which output panel each user reaches first and which sliders they treat as primary.

10 Comparison to existing tools

The mobility-based metapopulation epidemic literature is mature, and several production-grade systems implement variants of the same four-equation stack. What we add is not a new model class but a different point in the *interactivity-vs-rigour* trade space. Table 4 lays out where each comparable tool sits.

Tool	Strengths	Where this work differs
GLEAM / GLEAMviz (Northeastern MOBS Lab) [2]	Reference academic simulator: 3,300 sub-populations, stochastic compartmental dynamics, validated against multiple historical outbreaks. The model behind Chinazzi 2020 [4].	GLEAM is set-up-once, run-overnight; we are slider-driven and re-simulate in <1 s. We expose calibration (CRPS, log score, real backtest) and add an effective-distance polar projection and a particle-filter nowcast neither GLEAM nor GLEAMviz expose to non-experts.
EpiRisk (ISI Foundation)	Web-accessible Brockmann-style importation-risk lookup, used in professional response settings.	EpiRisk reports importation probabilities at fixed parameter settings; we let users move R_0, generation time, and intervention sliders and watch every output recompute, with cited bands and an LLM explanation.
WHO / CDC outbreak dashboards	Authoritative case-count reporting, surveillance integration, and trust.	Dashboards are post-hoc reporting tools, not forward simulators. They answer "what happened?"; we answer "what might?". The two are complementary, not competitive.
Imperial College COVID-19 Response Team reports	Highly cited bespoke per-outbreak modelling; sensitivity analyses; broad policy reach.	Imperial reports are bespoke per-outbreak runs published as PDFs; we are an interactive scenario-library generator with the same intervention decomposition (Tian 2020 [13]) at orders-of-magnitude faster turn-around.
Generic ML / GNN COVID forecasters [13]	Strong short-horizon accuracy on stationary signals; data-driven parameter learning.	ML methods do not extrapolate to *novel* pathogens (no historical data to learn from) and are difficult to audit. We deliberately use a structural model whose every term is a cited equation; ML lives only at the disease-parameter-RAG layer, not at the dynamics layer.

Table 4: Comparison to representative tools in the same problem space. Our contribution sits at the interactive end of the trade space, with the same underlying mathematical pipeline as the academic tools but exposed to non-experts in real time.

Three properties together separate this work from every entry in Table 4: **(i)** sub-second slider-driven re-simulation that scales the per-region Monte Carlo loop in vectorised numpy, **(ii)** a four-model ensemble whose membership is gated by a per-disease cryptic-pre-seeding flag drawn from the literature, and **(iii)** a particle-filter nowcast that lets the user plug in a few real case observations and watch the posterior R_0 and reporting fraction shift in roughly half a second. The first removes the expert-operator barrier of GLEAM; the second is, to our knowledge, novel; the third reframes the product from a teaching toy into a real-time forecasting harness.

9 Limitations and honest framing

We do not claim to predict real outbreaks. The system is a defensible scenario tool with literature-anchored parameters, calibrated against a specific real-world episode (COVID-19's first thirty days). Several known gaps deserve explicit mention.

Static mobility. The OD matrix is a snapshot. Rapti et al. 2022 demonstrate that time-varying mobility data measurably improves spatial forecasting; our reliance on a static prior is the simplest source of model–data mismatch and explains a large share of the backtest miss-rate.

Region resolution. Country-level aggregation hides intra-country heterogeneity. Within a few hours of work the resolution can be raised to the top five-hundred airports; for the demo we cap at seventy countries to keep the slider response under one second.

Reporting fraction. The cryptic-spread variant and the particle-filter nowcast both expose ρ, but the default forward simulation does not — it returns true infections, not reported cases. Davis et al. 2021 [6] document ρ as low as 0.01 for early COVID-19; users comparing model output to surveillance data should treat the model as an upper bound on observed counts unless they activate the nowcast and supply observed data.

No within-host or age structure. CFR is exposed as a slider but the SEIR compartments are population-aggregate. Pathogens with strong age-stratified transmission (e.g. measles) need an extension to the compartmental structure; we deliberately scoped the system to homogeneous populations, where the small model is more honest than a tunable bigger model would be.

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